

$$g(1)S(n) = \sum_{i=1}^n g(i)S(\lfloor \frac{n}{i} \rfloor) - \sum_{i=2}^n g(i)S(\lfloor \frac{n}{i} \rfloor) = \sum_{i=1}^n (f * g)(i) - \sum_{i=2}^n g(i)S(\lfloor \frac{n}{i} \rfloor)$$